

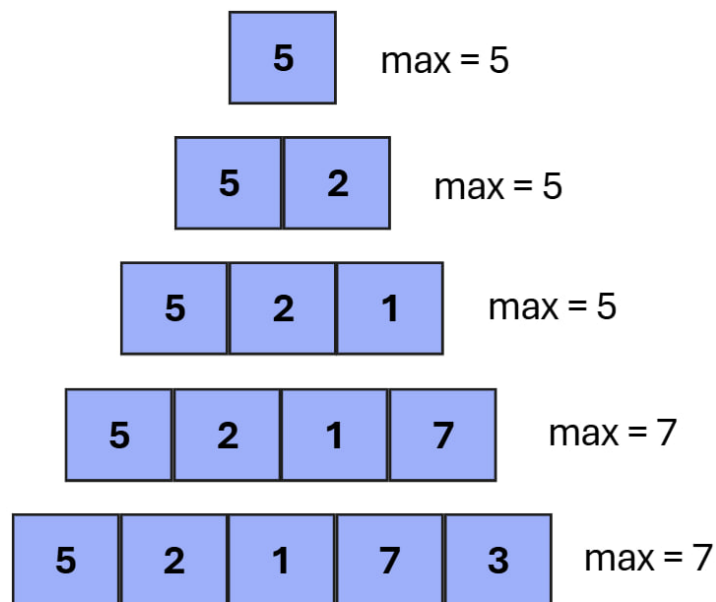
Peak Array

Input file: **standard input**
Output file: **standard output**
Time limit: **1 second**
Memory limit: **256 megabytes**

You are given an array A of N integers and Q queries.

1. For each position i , compute the maximum value among the first i elements of the array.
2. for each query x , print the maximum value among the first x values.

max computation when the array is [5, 2, 1, 7, 3]:



Input

The input consists of three lines:

- The first line of input consists of two integers N, Q ($1 \leq N, \leq 10^5$) ($1 \leq Q \leq 100$) the number of elements in the array and the number of queries.
- the second line contains N integers $A_1, A_2, \dots, A_N$ ($1 \leq A_i \leq 10^5$) the elements of the array.
- the third line contains q queries $x_1, x_2, \dots, x_q$ ($1 \leq x_i \leq N$) the positions you have to answer.

note that the positions in the queries are 1 based.

Output

Print a single line contains q space separated values $x_1, x_2, \dots, x_q$. the max value among the first x_i values.

Example

standard input	standard output
4 4 2 1 4 5 1 2 3 4	2 2 4 5